TOR AMPLITUDE DESCENDS ALONG A FAITHFULLY FLAT RING MAP

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Tor amplitude of a complex of modules over a ring. defined in the discussion of the same section, source lines 17100-17236. Supplies the property that has to be descended and the meaning of the interval [a, b].
- The derived tensor product computes the base change of the total tensor product when the ring map is flat. used at proof lines 17246-17250, the displayed identity of derived tensor products. Identifies the two sides whose cohomology is compared.
- A module over a ring is zero if and only if its base change along a faithfully flat map is zero. used at proof lines 17255-17258 to conclude that the Tor groups over R vanish outside the interval. Is the step that genuinely needs faithful flatness rather than flatness.

Source reference. Original lemma 068S: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8016–8016. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Lemma 0.2. *Let R be a ring. Let $a, b \in \mathbf{Z}$. Let $K^\bullet$ be a complex of R-modules. Let $R \to R'$ be a faithfully flat ring map. If the complex $K^\bullet \otimes_R R'$ has tor amplitude in $[a, b]$, then $K^\bullet$ has tor amplitude in $[a, b]$.*

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Let M be an R-module. Since $R \to R'$ is flat we see that

$$(M \otimes_R^{\mathbf{L}} K^\bullet) \otimes_R R' = ((M \otimes_R R') \otimes_{R'}^{\mathbf{L}} (K^\bullet \otimes_R R')$$

and taking cohomology commutes with tensoring with R'. Hence $\mathrm{Tor}_i^R(M, K^\bullet) \otimes_R R' = \mathrm{Tor}_i^{R'}(M \otimes_R R', K^\bullet \otimes_R R')$. Since $R \to R'$ is faithfully flat, the vanishing of $\mathrm{Tor}_i^{R'}(M \otimes_R R', K^\bullet \otimes_R R')$ for $i \notin [a, b]$ implies the same thing for $\mathrm{Tor}_i^R(M, K^\bullet)$. $\square$